

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1      | Dose 2     | Dose 3     | Dose 4     | Dose 5     | No OBD %    |
|-------------------------------------------------------------|-------------|------------|------------|------------|------------|-------------|
| <b>Scenario 1</b>                                           |             |            |            |            |            |             |
| DLT rate                                                    | 0.3         | 0.35       | 0.4        | 0.45       | 0.5        |             |
| Efficacy rate                                               | 0.05        | 0.06       | 0.07       | 0.08       | 0.09       |             |
| <b>OBD selection (%)</b>                                    |             |            |            |            |            |             |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>16.9</b> | <b>5.5</b> | <b>1.7</b> | <b>0.3</b> | <b>0.1</b> | <b>75.5</b> |
| New non-TITE   one-stage   alpha=0                          | 17.4        | 6.4        | 1.8        | 0.1        | 0.2        | 74.1        |
| New non-TITE   one-stage   alpha=0.3                        | 15.4        | 4.4        | 1          | 0.1        | 0.1        | 79          |
| New non-TITE   one-stage   alpha=0.6                        | 12.4        | 2.5        | 0.4        | 0          | 0          | 84.7        |
| New non-TITE   one-stage   alpha=0.9                        | 10.7        | 1.7        | 0.3        | 0          | 0          | 87.3        |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>17.6</b> | <b>5.7</b> | <b>0.9</b> | <b>0.4</b> | <b>0</b>   | <b>75.4</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 18          | 7.2        | 1.2        | 0          | 0.2        | 73.4        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 15.7        | 4.2        | 0.7        | 0          | 0.1        | 79.3        |
| New non-TITE   2-stage highest utility   alpha=0.6          | 13          | 2.2        | 0.3        | 0          | 0          | 84.5        |
| New non-TITE   2-stage highest utility   alpha=0.9          | 10.9        | 1.4        | 0.3        | 0          | 0          | 87.4        |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>16.6</b> | <b>5.7</b> | <b>1.5</b> | <b>0.5</b> | <b>0.1</b> | <b>75.6</b> |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 17.4        | 7.2        | 2          | 0.8        | 0.1        | 72.5        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 14.9        | 5.2        | 0.9        | 0.2        | 0          | 78.8        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 12.1        | 2.6        | 0.3        | 0          | 0          | 85          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 10.6        | 2.1        | 0          | 0          | 0          | 87.3        |
| <b>Mean administrations by dose</b>                         |             |            |            |            |            |             |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>11.1</b> | <b>4.3</b> | <b>2.2</b> | <b>0.8</b> | <b>0.2</b> |             |
| New non-TITE   one-stage   alpha=0                          | 11.1        | 4.5        | 2.2        | 0.9        | 0.3        |             |
| New non-TITE   one-stage   alpha=0.3                        | 11.5        | 3.9        | 1.7        | 0.6        | 0.2        |             |
| New non-TITE   one-stage   alpha=0.6                        | 11.7        | 3.3        | 1.2        | 0.3        | 0.1        |             |
| New non-TITE   one-stage   alpha=0.9                        | 12          | 2.6        | 0.8        | 0.2        | 0          |             |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>11.2</b> | <b>4.2</b> | <b>2.3</b> | <b>0.8</b> | <b>0.2</b> |             |
| New non-TITE   2-stage highest utility   alpha=0            | 11.1        | 4.4        | 2.3        | 1          | 0.3        |             |
| New non-TITE   2-stage highest utility   alpha=0.3          | 11.4        | 3.9        | 1.8        | 0.6        | 0.1        |             |
| New non-TITE   2-stage highest utility   alpha=0.6          | 11.7        | 3.3        | 1.3        | 0.3        | 0.1        |             |
| New non-TITE   2-stage highest utility   alpha=0.9          | 11.9        | 2.6        | 0.9        | 0.2        | 0          |             |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>11.1</b> | <b>4.2</b> | <b>2.2</b> | <b>0.9</b> | <b>0.2</b> |             |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 11          | 4.4        | 2.3        | 1          | 0.3        |             |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 11.3        | 3.9        | 1.7        | 0.6        | 0.1        |             |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 11.7        | 3.3        | 1.2        | 0.3        | 0.1        |             |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 11.9        | 2.7        | 0.9        | 0.2        | 0          |             |

Sources: cycle-2 new-design alpha grid and cycle-1 shared-additive oracle summary (both 1,000 simulations per populated row).

Oracle rows use Cycle\_Max = 1 and alpha = 0; No OBD % is the source's no-OBD selection percentage.

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1      | Dose 2      | Dose 3      | Dose 4      | Dose 5      | No OBD %   |
|-------------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|------------|
| <b>Scenario 16</b>                                          |             |             |             |             |             |            |
| DLT rate                                                    | 0.02        | 0.06        | 0.09        | 0.14        | 0.26        |            |
| Efficacy rate                                               | 0.2         | 0.2         | 0.2         | 0.2         | 0.2         |            |
| <b>OBD selection (%)</b>                                    |             |             |             |             |             |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>29</b>   | <b>21.8</b> | <b>19.9</b> | <b>18.9</b> | <b>10.4</b> | <b>0</b>   |
| New non-TITE   one-stage   alpha=0                          | 28.7        | 21.3        | 20.9        | 18.1        | 11          | 0          |
| New non-TITE   one-stage   alpha=0.3                        | 30.4        | 23.6        | 18.7        | 17.3        | 10          | 0          |
| New non-TITE   one-stage   alpha=0.6                        | 26.4        | 26          | 19.7        | 16.6        | 11.3        | 0          |
| New non-TITE   one-stage   alpha=0.9                        | 25.2        | 30.5        | 18.5        | 16.5        | 9.3         | 0          |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>32.5</b> | <b>22.7</b> | <b>16.3</b> | <b>17.8</b> | <b>10.6</b> | <b>0.1</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 34.1        | 21.3        | 16.6        | 16.4        | 11.5        | 0.1        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 30.9        | 26.8        | 18.6        | 16.3        | 7.4         | 0          |
| New non-TITE   2-stage highest utility   alpha=0.6          | 29.4        | 28.3        | 18          | 16.1        | 8.2         | 0          |
| New non-TITE   2-stage highest utility   alpha=0.9          | 27          | 33.9        | 16.9        | 15.6        | 6.6         | 0          |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>31.5</b> | <b>22.3</b> | <b>17.1</b> | <b>16.1</b> | <b>13</b>   | <b>0</b>   |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 33.9        | 22.2        | 16.5        | 15.4        | 12          | 0          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 32.1        | 23.8        | 17.6        | 15.8        | 10.7        | 0          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 29.5        | 26.5        | 18.9        | 14.8        | 10.3        | 0          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 26.4        | 29.8        | 17.6        | 18          | 8.2         | 0          |
| <b>Mean administrations by dose</b>                         |             |             |             |             |             |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>7.2</b>  | <b>6.7</b>  | <b>5.8</b>  | <b>5.8</b>  | <b>4.5</b>  |            |
| New non-TITE   one-stage   alpha=0                          | 7.3         | 6.6         | 5.7         | 5.7         | 4.7         |            |
| New non-TITE   one-stage   alpha=0.3                        | 7.3         | 7.1         | 5.7         | 5.6         | 4.3         |            |
| New non-TITE   one-stage   alpha=0.6                        | 7.2         | 7.5         | 5.8         | 5.5         | 4.1         |            |
| New non-TITE   one-stage   alpha=0.9                        | 7.2         | 8           | 5.9         | 5.3         | 3.6         |            |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>7.5</b>  | <b>6.9</b>  | <b>5.8</b>  | <b>5.6</b>  | <b>4.2</b>  |            |
| New non-TITE   2-stage highest utility   alpha=0            | 7.6         | 6.8         | 5.7         | 5.5         | 4.4         |            |
| New non-TITE   2-stage highest utility   alpha=0.3          | 7.4         | 7.3         | 5.7         | 5.5         | 4.1         |            |
| New non-TITE   2-stage highest utility   alpha=0.6          | 7.2         | 7.8         | 5.7         | 5.4         | 3.9         |            |
| New non-TITE   2-stage highest utility   alpha=0.9          | 7.2         | 8.2         | 5.5         | 5.3         | 3.7         |            |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>6.6</b>  | <b>6.3</b>  | <b>5.7</b>  | <b>6.1</b>  | <b>5.3</b>  |            |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 6.6         | 6.1         | 5.6         | 6.1         | 5.6         |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 6.6         | 6.5         | 5.7         | 6           | 5.2         |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 6.5         | 6.7         | 5.8         | 6           | 4.9         |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 6.5         | 7.1         | 5.8         | 6.1         | 4.6         |            |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1      | Dose 2      | Dose 3      | Dose 4      | Dose 5     | No OBD %   |
|-------------------------------------------------------------|-------------|-------------|-------------|-------------|------------|------------|
| <b>Scenario 20</b>                                          |             |             |             |             |            |            |
| DLT rate                                                    | 0.07        | 0.14        | 0.31        | 0.41        | 0.52       |            |
| Efficacy rate                                               | 0.05        | 0.25        | 0.25        | 0.25        | 0.25       |            |
| <b>OBD selection (%)</b>                                    |             |             |             |             |            |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>13.4</b> | <b>45.6</b> | <b>24.7</b> | <b>11.4</b> | <b>1.8</b> | <b>3.1</b> |
| New non-TITE   one-stage   alpha=0                          | 15.2        | 44          | 25.1        | 11.4        | 1.6        | 2.7        |
| New non-TITE   one-stage   alpha=0.3                        | 11.9        | 49.6        | 24.5        | 8.9         | 1.4        | 3.7        |
| New non-TITE   one-stage   alpha=0.6                        | 11          | 51.1        | 25.1        | 7.7         | 0.9        | 4.2        |
| New non-TITE   one-stage   alpha=0.9                        | 9.4         | 54.3        | 24.6        | 6.5         | 0.4        | 4.8        |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>13.5</b> | <b>51.2</b> | <b>24.4</b> | <b>6.6</b>  | <b>0.7</b> | <b>3.6</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 14.2        | 47.4        | 25.8        | 7.7         | 1.7        | 3.2        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 11.8        | 53.6        | 23.6        | 6.1         | 0.8        | 4.1        |
| New non-TITE   2-stage highest utility   alpha=0.6          | 10.2        | 55.4        | 24.2        | 5.4         | 0.4        | 4.4        |
| New non-TITE   2-stage highest utility   alpha=0.9          | 10          | 57          | 23.5        | 4.1         | 0.2        | 5.2        |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>9.2</b>  | <b>50</b>   | <b>25.3</b> | <b>9.8</b>  | <b>2</b>   | <b>3.7</b> |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 10.4        | 47.7        | 25.6        | 10.6        | 2.7        | 3          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 8.2         | 52.8        | 24.4        | 9.2         | 1.7        | 3.7        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 9.7         | 55.5        | 21.4        | 7.6         | 1.1        | 4.7        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 8.8         | 56.7        | 24.1        | 4.7         | 0.6        | 5.1        |
| <b>Mean administrations by dose</b>                         |             |             |             |             |            |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>5.3</b>  | <b>11.3</b> | <b>7.6</b>  | <b>4</b>    | <b>1.2</b> |            |
| New non-TITE   one-stage   alpha=0                          | 5.2         | 11.1        | 7.6         | 4.2         | 1.4        |            |
| New non-TITE   one-stage   alpha=0.3                        | 5.5         | 11.7        | 7.4         | 3.8         | 1          |            |
| New non-TITE   one-stage   alpha=0.6                        | 5.6         | 12.1        | 7.4         | 3.4         | 0.7        |            |
| New non-TITE   one-stage   alpha=0.9                        | 5.9         | 12.7        | 7.3         | 2.9         | 0.5        |            |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>5.4</b>  | <b>11.6</b> | <b>8</b>    | <b>3.5</b>  | <b>1</b>   |            |
| New non-TITE   2-stage highest utility   alpha=0            | 5.3         | 11.2        | 7.9         | 3.8         | 1.3        |            |
| New non-TITE   2-stage highest utility   alpha=0.3          | 5.6         | 11.6        | 7.8         | 3.3         | 1          |            |
| New non-TITE   2-stage highest utility   alpha=0.6          | 5.7         | 12.1        | 7.7         | 3           | 0.8        |            |
| New non-TITE   2-stage highest utility   alpha=0.9          | 6           | 12.5        | 7.4         | 2.7         | 0.7        |            |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>5.7</b>  | <b>9.8</b>  | <b>8</b>    | <b>4.4</b>  | <b>1.5</b> |            |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 5.7         | 9.5         | 8           | 4.6         | 1.8        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 6           | 9.9         | 8           | 4.2         | 1.3        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 6.3         | 10.3        | 7.9         | 3.8         | 1.1        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 6.7         | 10.6        | 7.9         | 3.3         | 0.8        |            |

Sources: cycle-2 new-design alpha grid and cycle-1 shared-additive oracle summary (both 1,000 simulations per populated row).

Oracle rows use Cycle\_Max = 1 and alpha = 0; No OBD % is the source's no-OBD selection percentage.

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1     | Dose 2     | Dose 3     | Dose 4      | Dose 5      | No OBD % |
|-------------------------------------------------------------|------------|------------|------------|-------------|-------------|----------|
| <b>Scenario 24</b>                                          |            |            |            |             |             |          |
| DLT rate                                                    | 0.01       | 0.02       | 0.05       | 0.1         | 0.25        |          |
| Efficacy rate                                               | 0.05       | 0.08       | 0.2        | 0.5         | 0.5         |          |
| <b>OBD selection (%)</b>                                    |            |            |            |             |             |          |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>1.3</b> | <b>1.4</b> | <b>5.2</b> | <b>58.8</b> | <b>33.3</b> | <b>0</b> |
| New non-TITE   one-stage   alpha=0                          | 1.4        | 1.4        | 4.7        | 59.1        | 33.4        | 0        |
| New non-TITE   one-stage   alpha=0.3                        | 1          | 0.9        | 3.3        | 62.3        | 32.5        | 0        |
| New non-TITE   one-stage   alpha=0.6                        | 0.4        | 0.6        | 3.4        | 65.1        | 30.5        | 0        |
| New non-TITE   one-stage   alpha=0.9                        | 0.5        | 0.6        | 3.3        | 67.6        | 28          | 0        |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>2</b>   | <b>1.1</b> | <b>6.8</b> | <b>56.7</b> | <b>33.4</b> | <b>0</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 2          | 1.5        | 6.2        | 55.5        | 34.8        | 0        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 1.1        | 1.4        | 5.5        | 55.9        | 36.1        | 0        |
| New non-TITE   2-stage highest utility   alpha=0.6          | 1.2        | 0.9        | 5.6        | 56.3        | 36          | 0        |
| New non-TITE   2-stage highest utility   alpha=0.9          | 0.8        | 0.8        | 5.3        | 57.3        | 35.8        | 0        |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>0.3</b> | <b>0.3</b> | <b>2.9</b> | <b>56.3</b> | <b>40.2</b> | <b>0</b> |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 0.4        | 0.8        | 3.6        | 56.6        | 38.6        | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 0.8        | 0.5        | 4          | 56.1        | 38.6        | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 0.6        | 0.6        | 4.2        | 54.6        | 40          | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 0.9        | 0.8        | 2.9        | 55.9        | 39.5        | 0        |
| <b>Mean administrations by dose</b>                         |            |            |            |             |             |          |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>3.4</b> | <b>3.5</b> | <b>4.5</b> | <b>12.5</b> | <b>6.2</b>  |          |
| New non-TITE   one-stage   alpha=0                          | 3.4        | 3.5        | 4.4        | 12.3        | 6.4         |          |
| New non-TITE   one-stage   alpha=0.3                        | 3.4        | 3.4        | 4.3        | 12.9        | 6           |          |
| New non-TITE   one-stage   alpha=0.6                        | 3.4        | 3.4        | 4.2        | 13.6        | 5.4         |          |
| New non-TITE   one-stage   alpha=0.9                        | 3.3        | 3.3        | 4.1        | 14          | 5.1         |          |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>3.4</b> | <b>3.5</b> | <b>4.4</b> | <b>10.9</b> | <b>7.7</b>  |          |
| New non-TITE   2-stage highest utility   alpha=0            | 3.4        | 3.5        | 4.4        | 10.6        | 8.1         |          |
| New non-TITE   2-stage highest utility   alpha=0.3          | 3.4        | 3.4        | 4.4        | 10.8        | 8           |          |
| New non-TITE   2-stage highest utility   alpha=0.6          | 3.4        | 3.4        | 4.2        | 11          | 8           |          |
| New non-TITE   2-stage highest utility   alpha=0.9          | 3.4        | 3.4        | 4.1        | 11.1        | 8           |          |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>3.6</b> | <b>3.8</b> | <b>4.9</b> | <b>9.5</b>  | <b>8.4</b>  |          |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 3.6        | 3.7        | 4.9        | 9.4         | 8.4         |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 3.5        | 3.7        | 4.8        | 9.4         | 8.5         |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 3.5        | 3.8        | 4.8        | 9.5         | 8.5         |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 3.5        | 3.8        | 4.8        | 9.5         | 8.4         |          |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1      | Dose 2      | Dose 3      | Dose 4      | Dose 5     | No OBD %   |
|-------------------------------------------------------------|-------------|-------------|-------------|-------------|------------|------------|
| <b>Scenario 27</b>                                          |             |             |             |             |            |            |
| DLT rate                                                    | 0.11        | 0.17        | 0.3         | 0.5         | 0.62       |            |
| Efficacy rate                                               | 0.1         | 0.2         | 0.4         | 0.42        | 0.45       |            |
| <b>OBD selection (%)</b>                                    |             |             |             |             |            |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>15.2</b> | <b>23.2</b> | <b>43</b>   | <b>11.2</b> | <b>0.6</b> | <b>6.8</b> |
| New non-TITE   one-stage   alpha=0                          | 16.3        | 21.2        | 44.7        | 11.3        | 0.7        | 5.8        |
| New non-TITE   one-stage   alpha=0.3                        | 15.9        | 23.1        | 45.7        | 7.9         | 0.5        | 6.9        |
| New non-TITE   one-stage   alpha=0.6                        | 13.8        | 28.7        | 44.5        | 6.1         | 0.2        | 6.7        |
| New non-TITE   one-stage   alpha=0.9                        | 16.3        | 33.3        | 39          | 3           | 0          | 8.4        |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>16.4</b> | <b>25.1</b> | <b>43.9</b> | <b>6.8</b>  | <b>0.6</b> | <b>7.2</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 16.3        | 23.9        | 44          | 8.4         | 0.9        | 6.5        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 17.8        | 27.6        | 41.5        | 5.4         | 0.4        | 7.3        |
| New non-TITE   2-stage highest utility   alpha=0.6          | 19.9        | 30.1        | 37.8        | 4.5         | 0.2        | 7.5        |
| New non-TITE   2-stage highest utility   alpha=0.9          | 22.1        | 34          | 32.4        | 3           | 0          | 8.5        |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>12</b>   | <b>24.5</b> | <b>45.2</b> | <b>10.6</b> | <b>1.1</b> | <b>6.6</b> |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 10.5        | 23.6        | 46.6        | 12.5        | 1.1        | 5.7        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 11.3        | 28.7        | 41.5        | 10.7        | 0.8        | 7          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 13.4        | 31.5        | 39.8        | 6.9         | 0.4        | 8          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 17.2        | 31.3        | 37.2        | 5.2         | 0.2        | 8.9        |
| <b>Mean administrations by dose</b>                         |             |             |             |             |            |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>7.3</b>  | <b>8.1</b>  | <b>9.7</b>  | <b>3.4</b>  | <b>0.6</b> |            |
| New non-TITE   one-stage   alpha=0                          | 7.1         | 7.9         | 10          | 3.6         | 0.7        |            |
| New non-TITE   one-stage   alpha=0.3                        | 7.5         | 8.3         | 9.8         | 2.9         | 0.4        |            |
| New non-TITE   one-stage   alpha=0.6                        | 7.9         | 8.8         | 9.6         | 2.5         | 0.3        |            |
| New non-TITE   one-stage   alpha=0.9                        | 8.4         | 9.4         | 9           | 1.9         | 0.1        |            |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>7.8</b>  | <b>8.2</b>  | <b>9.4</b>  | <b>3</b>    | <b>0.7</b> |            |
| New non-TITE   2-stage highest utility   alpha=0            | 7.7         | 7.9         | 9.4         | 3.2         | 0.9        |            |
| New non-TITE   2-stage highest utility   alpha=0.3          | 8.3         | 8.3         | 8.9         | 2.8         | 0.7        |            |
| New non-TITE   2-stage highest utility   alpha=0.6          | 8.7         | 8.8         | 8.6         | 2.5         | 0.5        |            |
| New non-TITE   2-stage highest utility   alpha=0.9          | 9.3         | 9.3         | 7.9         | 2.1         | 0.4        |            |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>7.6</b>  | <b>8.4</b>  | <b>8.5</b>  | <b>3.7</b>  | <b>0.9</b> |            |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 7.4         | 8.2         | 8.4         | 4           | 1.1        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 8           | 8.6         | 8.1         | 3.6         | 0.8        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 8.5         | 8.9         | 7.9         | 3.1         | 0.6        |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 9.3         | 9.2         | 7.5         | 2.5         | 0.4        |            |

Sources: cycle-2 new-design alpha grid and cycle-1 shared-additive oracle summary (both 1,000 simulations per populated row).

Oracle rows use Cycle\_Max = 1 and alpha = 0; No OBD % is the source's no-OBD selection percentage.

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                             | Dose 1      | Dose 2      | Dose 3     | Dose 4     | Dose 5      | No OBD %   |
|-------------------------------------------------------------|-------------|-------------|------------|------------|-------------|------------|
| <b>Scenario 38</b>                                          |             |             |            |            |             |            |
| DLT rate                                                    | 0.05        | 0.1         | 0.12       | 0.15       | 0.3         |            |
| Efficacy rate                                               | 0.05        | 0.08        | 0.12       | 0.15       | 0.5         |            |
| <b>OBD selection (%)</b>                                    |             |             |            |            |             |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>14.1</b> | <b>7.5</b>  | <b>7.1</b> | <b>7.5</b> | <b>62.4</b> | <b>1.4</b> |
| New non-TITE   one-stage   alpha=0                          | 13.8        | 7.2         | 6.8        | 7.8        | 63.5        | 0.9        |
| New non-TITE   one-stage   alpha=0.3                        | 15          | 7.5         | 7.4        | 11.3       | 57.6        | 1.2        |
| New non-TITE   one-stage   alpha=0.6                        | 15.6        | 9.3         | 8.2        | 11.2       | 54          | 1.7        |
| New non-TITE   one-stage   alpha=0.9                        | 15.6        | 8.9         | 9.5        | 13.2       | 51          | 1.8        |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>19.2</b> | <b>10.3</b> | <b>8.6</b> | <b>7.4</b> | <b>52.8</b> | <b>1.7</b> |
| New non-TITE   2-stage highest utility   alpha=0            | 17.9        | 10          | 7.3        | 7.5        | 56          | 1.3        |
| New non-TITE   2-stage highest utility   alpha=0.3          | 18.5        | 10.7        | 8.9        | 10         | 50          | 1.9        |
| New non-TITE   2-stage highest utility   alpha=0.6          | 20.3        | 12.4        | 8.7        | 10.7       | 45.7        | 2.2        |
| New non-TITE   2-stage highest utility   alpha=0.9          | 20.6        | 12.6        | 12         | 12.1       | 40.5        | 2.2        |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>12</b>   | <b>5.6</b>  | <b>7.4</b> | <b>7.8</b> | <b>65.8</b> | <b>1.4</b> |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 11.7        | 6           | 5.5        | 8.5        | 67.3        | 1          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 12.3        | 7.5         | 8.3        | 10.6       | 59.8        | 1.5        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 12.8        | 8.6         | 8.2        | 11.2       | 57          | 2.2        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 11.9        | 10          | 11.7       | 11.1       | 53.2        | 2.1        |
| <b>Mean administrations by dose</b>                         |             |             |            |            |             |            |
| <b>Oracle   one-stage   cycle 1, alpha=0</b>                | <b>4.6</b>  | <b>4.6</b>  | <b>5.1</b> | <b>5.4</b> | <b>10.1</b> |            |
| New non-TITE   one-stage   alpha=0                          | 4.6         | 4.6         | 5          | 5.3        | 10.4        |            |
| New non-TITE   one-stage   alpha=0.3                        | 4.7         | 4.8         | 5.2        | 5.5        | 9.5         |            |
| New non-TITE   one-stage   alpha=0.6                        | 4.8         | 5           | 5.4        | 5.5        | 9           |            |
| New non-TITE   one-stage   alpha=0.9                        | 4.8         | 5.3         | 5.6        | 5.6        | 8.4         |            |
| <b>Oracle   2-stage highest utility   cycle 1, alpha=0</b>  | <b>4.9</b>  | <b>4.8</b>  | <b>5.5</b> | <b>5.6</b> | <b>9.1</b>  |            |
| New non-TITE   2-stage highest utility   alpha=0            | 4.8         | 4.7         | 5.3        | 5.4        | 9.6         |            |
| New non-TITE   2-stage highest utility   alpha=0.3          | 5           | 4.9         | 5.6        | 5.7        | 8.6         |            |
| New non-TITE   2-stage highest utility   alpha=0.6          | 5.1         | 5.2         | 5.8        | 5.7        | 8           |            |
| New non-TITE   2-stage highest utility   alpha=0.9          | 5.2         | 5.6         | 6          | 5.7        | 7.2         |            |
| <b>Oracle   2-stage top-2 randomized   cycle 1, alpha=0</b> | <b>4.8</b>  | <b>4.9</b>  | <b>5.4</b> | <b>6.8</b> | <b>7.8</b>  |            |
| New non-TITE   2-stage top-2 randomized   alpha=0           | 4.7         | 4.8         | 5.3        | 6.9        | 8.2         |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.3         | 4.8         | 5           | 5.5        | 6.8        | 7.5         |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.6         | 4.9         | 5.4         | 5.7        | 6.7        | 7           |            |
| New non-TITE   2-stage top-2 randomized   alpha=0.9         | 5           | 5.7         | 5.9        | 6.6        | 6.5         |            |

Sources: cycle-2 new-design alpha grid and cycle-1 shared-additive oracle summary (both 1,000 simulations per populated row).

Oracle rows use Cycle\_Max = 1 and alpha = 0; No OBD % is the source's no-OBD selection percentage.